Reading the Biopsy: Paediatric Gastrointestinal Histology for the Gastroenterologist.

Imagine three pathology reports arriving on the same afternoon. The first says that a biopsy from the caecum of a ten-year-old with abdominal pain contains forty-five eosinophils per high-power field, and the referring doctor is already planning an elimination diet for eosinophilic colitis. The second describes an eight-year-old with bloody diarrhoea, inflammation throughout the colon, and a rectum that looks almost normal, and the words "rectal sparing, consider Crohn's disease" appear in the comment. The third describes colitis with crypt cell death and almost no plasma cells, in a child who has had recurrent pneumonia. In the first case, the count is normal for that part of the bowel. In the second, rectal sparing is a well-known feature of ulcerative colitis in young children. In the third, the missing plasma cells are the most important finding on the slide, because they point to an immunodeficiency. Each report is correct. Each can lead to the wrong decision if the gastroenterologist does not know how to read it.

The clinical question of this chapter is how to read a paediatric gastrointestinal biopsy report as a clinician. The main idea is this. Normal values in the gut depend on the segment, chronic inflammation leaves a sequence of features that appear at different speeds, granulomas and dying crypt cells have several causes besides Crohn's disease, and coeliac histology depends on orientation and on who reads it. A biopsy answers a clinical question only if the question is asked, so the request form should state the clinical problem, the medicines the child is taking, any transplant or immunodeficiency, and the bowel preparation used.

We will move through five topics. First, eosinophils: what is normal in each segment, and the thresholds for disease. Second, chronic versus acute colitis, and why children break the adult rules. Third, granulomas. Fourth, apoptosis, meaning programmed cell death in the crypts, in graft-versus-host disease, drug injury and immunodeficiency. Fifth, coeliac disease and the Marsh classification. We then compare the North American and European documents and finish with what to do differently.

A note on sources. For normal eosinophil counts we use the 2022 study of healthy children by Hoofien and colleagues for the eosinophilic disease working group of the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN, the 2014 review of diagnostic thresholds by Margaret Collins, and the 2024 joint guideline of ESPGHAN and the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, on eosinophilic disorders beyond the oesophagus, led by Papadopoulou. For colitis we use the 2013 European consensus on the histopathology of inflammatory bowel disease from the European Crohn's and Colitis Organisation and the European Society of Pathology, known as ECCO and ESP, led by Magro, and the 2014 revised Porto criteria of ESPGHAN, led by Levine. For coeliac disease we use the 2020 ESPGHAN guideline led by Husby and the 2023 guideline of the American College of Gastroenterology, the ACG, led by Rubio-Tapia.

Part one. Each feature is a clock.

Before looking at numbers, it helps to understand what each feature on a slide means biologically, because each one appears on its own time scale. Basal plasmacytosis means an increased number of plasma cells crowded between the bottom of the crypts and the thin muscle layer beneath the mucosa. It reflects a chronic, antigen-driven immune response, and it is the earliest feature that favours inflammatory bowel disease over infection, appearing within days. Crypt architectural distortion means crypts that are branched, shortened, irregularly spaced, or no longer parallel. It reflects repeated cycles of crypt destruction and regeneration, and it takes weeks to months to develop. Paneth cell metaplasia, meaning the appearance of Paneth cells in the left colon where they are not normally found, reflects long-standing repair, and is a marker of chronicity rather than a way to tell one disease from another. A non-necrotising granuloma, a tight cluster of activated macrophages without dead tissue in the centre, reflects a macrophage response to a persistent antigen. Crypt apoptosis, single dying cells at the base of the crypts, reflects cytotoxic or drug injury to the stem cell zone and can appear within days. And in the small bowel, raised intraepithelial lymphocytes and villous atrophy reflect a T-cell response to gluten, with enterocytes being lost faster than the crypts can replace them.

By contrast, acute self-limited colitis, which is usually infectious, keeps a normal crypt architecture. The neutrophils sit high in the lamina propria, near the surface, and there is oedema but no basal plasma cells. The classic study by Surawicz and colleagues in Gastroenterology in 1994 compared acute self-limited colitis with new-onset inflammatory bowel disease and showed that chronic changes could already be seen as early as seven days after symptoms began.

Part two. Eosinophils: normal is not zero, and not one number.

The most useful modern data on normal eosinophils come from the 2022 ESPGHAN study of 202 healthy children, median age thirteen, from centres in Italy, the United Kingdom and Israel, all of whom had a normal endoscopy and at least a year of follow-up without organic disease. The counts were expressed as eosinophils per square millimetre. The median was zero in the oesophagus and zero in the stomach. It was twenty in the duodenal bulb and twenty in the second part of the duodenum, twenty-nine in the terminal ileum, fifty-three in the caecum and fifty-five in the ascending colon. From there it fell steadily: thirty-eight in the transverse colon, twenty-nine in the descending colon, twenty-five in the sigmoid and only thirteen in the rectum. The ranges were wide, reaching 232 per square millimetre in the caecum and 247 in the ileum of healthy children. Counts differed significantly between the three centres, but they did not differ by age group or by functional diagnosis.

Three lessons follow. First, eosinophil density falls from right to left across the colon, so the same count can be ordinary in the caecum and high in the rectum. Second, geography matters, so a single universal cut-off is an approximation. Third, the oesophagus is different: the normal median is zero, so any eosinophil in the squamous epithelium deserves an explanation.

How are disease thresholds set? In her 2014 review, Collins proposed that a count of about twice the normal maximum should be considered abnormal. Using older normal data, the normal maximum per high-power field was about twenty-six in the duodenum, twenty-eight in the ileum,

fifty in the caecum and ascending colon, forty-two in the transverse and descending colon and thirty-two in the rectosigmoid. Her thresholds were therefore fifty-two in the duodenum, fifty-six in the ileum, one hundred in the right colon, eighty-four in the transverse and descending colon, and sixty-four in the rectosigmoid. For the stomach she proposed thirty or more per field in at least five fields, and seventy or more in at least three. The 2024 joint ESPGHAN and NASPGHAN guideline rounded these to thirty in the stomach, fifty in the duodenum, sixty in the ileum, one hundred in the caecum and ascending colon, eighty in the transverse and descending colon, and sixty in the rectosigmoid. One important caution: adult clinical trials have used a much lower duodenal threshold, thirty or more per field in three fields, so a duodenal count of thirty-five can be called eosinophilic duodenitis in an adult trial but would be normal under the paediatric guideline. And a count is never enough on its own; it needs symptoms and the exclusion of parasites, drugs, inflammatory bowel disease and coeliac disease.

Part three. Say the segment, count the area.

For eosinophilic oesophagitis, the 2018 international AGREE consensus, led by Dellon, requires a peak count of fifteen or more eosinophils per high-power field, which it equates to about sixty per square millimetre, together with symptoms of oesophageal dysfunction and exclusion of other causes. A trial of a proton pump inhibitor is no longer required for diagnosis.

In 2022 an international Delphi process of 91 experts, published by Dellon and colleagues in Clinical Gastroenterology and Hepatology, agreed a new nomenclature. The umbrella term is eosinophilic gastrointestinal disease. Each segment has its own name: eosinophilic gastritis, eosinophilic enteritis, which includes duodenitis, jejunitis and ileitis, and eosinophilic colitis. Ninety-six percent agreed to stop using "eosinophilic gastroenteritis" as the umbrella term, and when more than one site is involved, every site should be named.

There is one technical trap. The area of a high-power field depends on the microscope, and studies have used fields of quite different sizes. The AGREE equivalence of fifteen per field and sixty per square millimetre implies a field of about a quarter of a square millimetre. A wider field counts more cells in the same tissue, so the same biopsy can cross or miss a threshold depending on the microscope. The practical solution is to ask the pathologist for density per square millimetre, the number of fields above threshold, and the accompanying features: eosinophils within crypt epithelium, eosinophilic crypt abscesses, sheets of eosinophils in the lamina propria, and degranulation.

Part four. Chronic or self-limited colitis?

When a child has bloody diarrhoea, the first histological question is whether the colitis is chronic, pointing to inflammatory bowel disease, or acute and self-limited, usually infectious. The features that favour chronicity fall into three groups. In the architecture, look for crypt branching, crypt shortening and loss of the normal parallel arrangement; but crypt atrophy alone should not be counted as distortion, and poor orientation of the biopsy can mimic it. In the lamina propria, look for basal plasmacytosis, basal lymphoid aggregates and an increase in cells throughout the full

thickness of the mucosa rather than just near the surface. In the epithelium, look for Paneth cell metaplasia in the left colon and mucin depletion. Cryptitis and crypt abscesses also favour inflammatory bowel disease, though they occur in infection as well. The ECCO and ESP consensus recommends at least two biopsies from each of at least five sites, including the ileum and rectum, each site in a separate labelled container, and it notes that segmental biopsies raised diagnostic accuracy from sixty-six to ninety-two percent. A British reporting guideline adds that two or three neutrophils within the epithelium of a biopsy can simply be an effect of the bowel preparation.

Timing is crucial. In data summarised in the ECCO and ESP consensus, basal plasmacytosis was present in sixty-three percent of first attacks of ulcerative colitis and sixty-two percent of Crohn's disease, but in only six percent of infectious colitis. Within two weeks of symptom onset, basal plasmacytosis was present in thirty-eight percent of patients who turned out to have inflammatory bowel disease, but crypt distortion was present in only about twenty percent. So plasma cells come before architecture. If the first biopsy is very early and equivocal, the right course is to complete the infectious work-up and repeat the biopsy if symptoms persist, rather than to declare that inflammatory bowel disease has been excluded.

Part five. Children break the adult rules.

Classical teaching says that ulcerative colitis is continuous from the rectum upward and shows chronic changes. Children often break both rules. In a 2004 study in the American Journal of Surgical Pathology, Glickman and colleagues compared seventy-three children with thirty-eight adults with new, untreated ulcerative colitis. Patchy chronic changes were seen in twenty-one percent of children and in no adults. Relative rectal sparing was seen in twenty-three percent of children and in no adults. Little or no crypt distortion on the rectal biopsy was seen in eight percent of children and in no adults. And subtotal or total colitis was found in forty-two percent of children compared with eleven percent of adults. A companion study by Robert and colleagues showed that these differences are concentrated in children younger than ten, who had less crypt branching, fewer plasma cells and less cryptitis than adults, whereas adolescents looked like adults. The ECCO and ESP consensus similarly reports basal plasmacytosis in fifty-eight percent of children with ulcerative colitis, patchiness in twenty-one percent and rectal sparing in thirty percent.

The 2014 revised Porto criteria build this into the diagnosis. They require upper endoscopy and ileocolonoscopy in every child with suspected inflammatory bowel disease, and they define atypical phenotypes that should still be classified as ulcerative colitis: rectal sparing, short duration of disease without chronic changes, a caecal patch of inflammation, upper gastrointestinal erosions or ulcers without granulomas, and acute severe colitis. The rule for practice is simple: in a young child, a normal rectum or the absence of crypt distortion does not rule out ulcerative colitis, and rectal sparing alone is not a reason to relabel the disease as Crohn's.

Part six. Granulomas: look high and look early.

Granulomas are the histological hallmark of Crohn's disease, and they are common in children if they are looked for in the right place and at the right time. In a 2008 study of 282 children with both upper and lower endoscopy, De Matos and colleagues found granulomas in sixty-one percent of untreated children at diagnosis, but in only about twenty-five percent of children who had already received treatment. In forty-two percent of children with granulomas, they were found only in upper gastrointestinal or terminal ileal biopsies. Granulomas were associated with perianal disease and with more extensive upper tract involvement. In a 2007 study by Rubio and colleagues, granulomas were found at the first colonoscopy in sixty-seven percent of children and sixty-six percent of adults, but at later colonoscopies they persisted in fifty-four percent of children compared with only eighteen percent of adults. In a 2017 Israeli cohort of 289 children, Rothschild and colleagues found granulomas in thirty-four percent, and those children had more upper tract disease and needed biologic therapy more often. The two practical lessons are to biopsy the upper tract and the ileum every time, and to biopsy before starting steroids if possible.

Not every granuloma is Crohn's disease. A Crohn's granuloma is non-necrotising, made of epithelioid macrophages, often small and few, and sits in the lamina propria away from the crypts. A crypt rupture granuloma, made of macrophages gathered around spilled mucin or a broken crypt, is a reaction to crypt damage, occurs in ulcerative colitis and in infection, and is not diagnostic. In chronic granulomatous disease, an inherited failure of neutrophil and macrophage killing, the granulomas are often confluent and accompanied by pigment-laden macrophages, and a neutrophil oxidative burst test should be part of the work-up of very early onset disease. Granulomas with necrosis or caseation should prompt a search for tuberculosis, Yersinia or Actinomyces, with stains and cultures. Foreign body granulomas contain refractile material, and granulomas can also occur in common variable immunodeficiency.

In the stomach, focally enhanced gastritis, meaning small focal collections of inflammatory cells around a few glands without Helicobacter pylori, is seen in forty-three to seventy-six percent of children with Crohn's disease, but also in eight to twenty-one percent of children with ulcerative colitis. It supports a diagnosis of inflammatory bowel disease, but it does not separate Crohn's disease from ulcerative colitis. Indeed, upper gastrointestinal changes occur in up to three quarters of children with ulcerative colitis.

Part seven. Apoptosis: counting dead cells at the crypt base.

Apoptosis is programmed cell death, and on a biopsy it appears as small round fragments of nuclear debris within the crypt epithelium. Its classic setting is graft-versus-host disease after a stem cell transplant, in which donor T cells attack the recipient's crypt stem cells. The traditional grading, often attributed to Lerner, has four grades. In grade one there is crypt apoptosis without crypt loss. In grade two individual crypts are lost. In grade three two or more neighbouring crypts are lost. And in grade four there is diffuse crypt loss with denuded mucosa. The 2014 consensus of the United States National Institutes of Health, the NIH, published in 2015, sets the minimum criterion for the diagnosis as at least one apoptotic crypt cell per biopsy fragment, after the pathologist has reviewed several serial levels. A 2013 study by Lin and colleagues found that more

than six apoptotic bodies per ten contiguous crypts is definitive, whereas six or fewer is indeterminate. The histological grade, however, correlates poorly with clinical severity.

Many other causes produce the same picture. Bowel preparation, especially oral sodium phosphate, can cause basal crypt apoptosis and small aphthous lesions, so it is worth knowing which preparation was used. Mycophenolate mofetil, an immunosuppressant used after organ transplantation and in autoimmune disease, is a particular challenge. In a 2009 study by Selbst and colleagues, the colonic biopsies of patients with mycophenolate-associated colitis were normal or near normal in thirty-one percent, looked like inflammatory bowel disease in twenty-eight percent, like graft-versus-host disease in nineteen percent, like acute self-limited colitis in sixteen percent and like ischaemia in three percent. A 2013 series by Lee and colleagues found an average of twenty-six and a half apoptotic bodies per hundred crypts. A 2013 study by Star and colleagues found that more than fifteen eosinophils in the lamina propria per ten high-power fields favoured mycophenolate over graft-versus-host disease. In a transplant child, graft-versus-host disease, cytomegalovirus and mycophenolate injury can coexist, so the pathologist needs the drug list and a cytomegalovirus stain, and improvement after dose reduction is often the practical proof.

The immune system offers further mimics. In common variable immunodeficiency, which can cause an enteropathy in a substantial minority of patients, plasma cells are absent or very scarce in about two thirds of biopsies, crypt apoptosis resembling graft-versus-host disease is seen in about a fifth, and there may be coeliac-like villous blunting, lymphoid hyperplasia and Giardia. In very early onset inflammatory bowel disease, an apoptotic pattern with crypt dropout suggests monogenic disease such as defects in the TTC7A or XIAP genes, while defects in the interleukin 10 receptor produce a Crohn-like picture; genetic testing should come before biologic therapy. In adults treated with checkpoint inhibitor immunotherapy, colitis typically shows active inflammation with prominent crypt apoptosis but without basal plasmacytosis or chronic architecture, and the drug idelalisib, which inhibits the delta form of the enzyme phosphoinositide 3-kinase, causes raised intraepithelial lymphocytes, crypt apoptosis and cryptitis. The clinical lesson is that an "inflammatory bowel disease" biopsy without plasma cells is a clue, not a curiosity. Measure the immunoglobulins. Coeliac-like atrophy in common variable immunodeficiency does not respond to gluten withdrawal, and coeliac serology cannot be interpreted when the child does not make antibodies.

Part eight. Coeliac histology: the Marsh classification and its simpler heir.

The duodenal lesion of coeliac disease is graded with the Marsh classification as modified by Oberhuber in 1999. Grade zero is normal. Grade one has raised intraepithelial lymphocytes with normal crypts and villi. Grade two adds crypt hyperplasia, still with normal villi. Grade three adds villous atrophy and is subdivided: three a is mild or partial atrophy, three b is marked or subtotal atrophy, and three c is total atrophy with a flat mucosa. Oberhuber's original cut-off for raised intraepithelial lymphocytes was forty per hundred epithelial cells. The modern cut-off of twenty-five per hundred comes from a 2002 study by Hayat and colleagues in the Journal of Clinical Pathology, which found a normal mean of eleven with a range from about two to

twenty-six, and a mean ratio of villus height to crypt depth of 1.82. A simpler scheme proposed by Corazza and Villanacci collapses these grades into three: grade A, non-atrophic, with raised lymphocytes; grade B1, atrophic but with villi still detectable; and grade B2, flat, with no detectable villi.

The simpler scheme exists because the Marsh grade depends heavily on who reads the slide. In a 2007 study by Corazza and colleagues, six pathologists grading sixty slides twice achieved a mean kappa, a measure of agreement beyond chance where one is perfect, of only 0.35 with the Marsh and Oberhuber scheme, compared with 0.55 with the simpler scheme. In a 2012 study by Arguelles-Grande and colleagues, slides referred to a tertiary centre were re-read by an expert. Agreement with the expert was 0.42 for commercial laboratories, 0.47 for community hospitals and 0.89 for university hospitals. The diagnosis changed in a quarter of cases, two thirds of the original reports lacked a Marsh grade, and in forty-three percent of cases only one well-oriented piece was available. A 2008 Italian multicentre study by Bonamico and colleagues found that 2.4 percent of 665 children with coeliac disease had atrophy only in the duodenal bulb, which is why a bulb biopsy is part of the protocol.

The 2020 ESPGHAN guideline requires at least four biopsies from the distal duodenum and at least one from the bulb, taken while the child is eating gluten, and it states that Marsh grade two or three is needed to confirm coeliac disease; Marsh grade one alone is insufficient. It asks for optimally oriented biopsies and treats a ratio of villus height to crypt depth below two as a lesion. It also allows diagnosis without biopsy in children whose tissue transglutaminase IgA antibody is at least ten times the upper limit of normal, with a positive endomysial antibody on a second blood sample.

Part nine. What the guidelines say.

On the coeliac biopsy protocol, North America and Europe are aligned: the 2023 ACG guideline also asks for bulb biopsies plus at least four distal duodenal biopsies. On the histology required, they are aligned: the ACG calls raised lymphocytes without atrophy non-specific, and ESPGHAN requires Marsh two or three. On the no-biopsy route there is a material difference. ESPGHAN adopted it first and more firmly for children, while the ACG accepts it for children only as a conditional recommendation and does not endorse it for adults. That matters at transition, when a teenager diagnosed without a biopsy in Europe may meet adult North American services that expect one. On eosinophilic oesophagitis, both sides use the AGREE criteria and are aligned. For eosinophilic disease beyond the oesophagus, paediatric thresholds come from one joint document, but they differ from the much lower thresholds used in adult trials. And for inflammatory bowel disease histology, Europe has the dedicated ECCO and ESP consensus and the Porto definitions of atypical ulcerative colitis, whereas North America has no equivalent paediatric histology standard.

Part ten. What to do differently.

Let us return to the three reports and add two more. Forty-five eosinophils per field in the caecum of a child with abdominal pain is within the normal range for the right colon; do not label

eosinophilic colitis, and look elsewhere for the cause of the pain. Acute colitis with preserved architecture on the fifth day of bloody diarrhoea suggests infection, but inflammatory bowel disease cannot be excluded this early; send the stool tests and re-biopsy if symptoms persist. Pancolitis with rectal sparing in an eight-year-old without granulomas is atypical paediatric ulcerative colitis under the Porto criteria; do not relabel it Crohn's disease on the rectum alone, and check the upper tract biopsies. Colitis with crypt apoptosis and no plasma cells should prompt immunoglobulins, lymphocyte subsets and a genetic panel for common variable immunodeficiency or monogenic very early onset disease. And a duodenum with thirty-two intraepithelial lymphocytes per hundred, normal villi and a positive tissue transglutaminase is Marsh grade one, which is not diagnostic; check the orientation and the bulb biopsy, keep the child on gluten, repeat the serology, and do not attach a lifelong label on Marsh one alone.

The whole chapter can be held in a few sentences. Normal depends on the segment, so ask for density and compare with the right reference. Plasma cells come before architecture, and children may show neither. Biopsy high, biopsy the ileum, and biopsy before steroids if you want to find granulomas. When the crypts are dying, ask about drugs, transplants and immunity. And when coeliac histology decides a diagnosis, make sure the pieces are well oriented and that an experienced pathologist reads them.

Questions to test yourself.

Question one. Where in the healthy colon is eosinophil density highest, and where is it lowest? Answer: highest in the caecum and ascending colon, with medians of fifty-three and fifty-five per square millimetre, and lowest in the rectum, with a median of thirteen.

Question two. What are the joint 2024 paediatric eosinophil thresholds per high-power field? Answer: thirty in the stomach, fifty in the duodenum, sixty in the ileum, one hundred in the caecum and ascending colon, eighty in the transverse and descending colon, and sixty in the rectosigmoid.

Question three. Which histological feature appears earliest in inflammatory bowel disease, and how often is it seen in infectious colitis? Answer: basal plasmacytosis, seen in about sixty-three percent of first attacks of ulcerative colitis but only six percent of infectious colitis.

Question four. Name three ways in which new-onset ulcerative colitis in children differs histologically from adults. Answer: patchy chronicity, relative rectal sparing and minimal crypt distortion, each seen in a minority of children and in no adults in the Glickman study, together with more extensive disease.

Question five. Why should a crypt rupture granuloma not be used to diagnose Crohn's disease? Answer: because it is a reaction to crypt damage and spilled mucin, seen also in ulcerative colitis and infection; only lamina propria granulomas away from crypts support Crohn's disease.

Question six. What is the NIH minimum histological criterion for gastrointestinal graft-versus-host disease? Answer: at least one apoptotic crypt cell per biopsy fragment, after review of serial levels.

Question seven. What should an inflammatory colitis without plasma cells make you think of? Answer: common variable immunodeficiency, in which plasma cells are absent or scarce in about two thirds of cases, and monogenic very early onset inflammatory bowel disease.

Question eight. How reproducible is the Marsh and Oberhuber classification? Answer: poorly; the mean kappa was 0.35 in one study, compared with 0.55 for the simpler Corazza and Villanacci scheme, and agreement with an expert ranged from 0.42 for commercial laboratories to 0.89 for university hospitals.

End of chapter.